



The advantage of HDFS is that it is scalable, i.e., any number of systems can be added at any point in time. It works on commodity hardware, so it is easy to keep costs low as compared to other databases. HDFS is mainly designed for large files, and it works on the concept of write once and read many times. In HDFS, individual files are broken into blocks of fixed size (typically 64MB) and stored across a cluster of nodes (not necessarily on the same machine). These files can be more than the size of an individual machine's hard drive. The individual machines are called data nodes.

- Virtual Machine software which can be downloaded from <http://www.vmware.com/support/download-player>
- Hadoop VM from Cloudera <http://content.udacity-data.com/courses/ud617/Cloudera-Udacity-Training-VM-4.1.1.c.zip>
- Hadoop has introduced several versions of the VM. The image present in the following link is 0.18 version of Hadoop <http://developer.yahoo.com/hadoop/tutorial/module3.html>
- Putty can be downloaded from <http://www.putty.org/>
- The last is WinScp and this can be downloaded from <http://winscp.net/eng/download.php>

Now, in order to interact with the machine, an SSH connection should be established; so in a terminal, type the following commands. First install the client, then the server. The SSH key will be generated by this and can be shared with other machines in the cluster to get the connection.



The screenshot shows a Notepad++ window titled "profile (.bat) - gedit". The menu bar includes File, Edit, View, Search, Tools, Documents, and Help. The toolbar contains icons for Open, Save, Undo, and other standard editing functions. The "profile" tab is active, displaying the following shell script content:

```
#!/bin/bash

setenv() {
    if [ -d "$1" ]; then
        echo "$1"
    fi
}

PATH=$PATH:$HOME/bin
LD_LIBRARY_PATH=$LD_LIBRARY_PATH:$HOME/bin
export PATH LD_LIBRARY_PATH

export JAVA_HOME=C:\Program Files\Java\jdk1.7.0_51
export PATH=$PATH:$HOME/bin
```

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